

Junaid Ahmed Mohammed

Senior Data Engineer | M.S. Applied Data Science

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PROFESSIONAL EXPERIENCE

INDIANA UNIVERSITY INDIANAPOLIS

Indianapolis, Indiana

Graduate Research Assistant — COMET Lab

Jan 2024 – Dec 2025

- **Engineered** end-to-end streaming ML pipeline: Kafka ingestion, PySpark Structured Streaming with multi-rule validation (revenue, cost, date, categorical logic), and PostgreSQL load. Orchestrated via **Airflow DAG** with 50,000-row threshold trigger, deduplication, and Spark-submit model training for revenue and units-sold forecasting. Containerized via Docker Compose.
- **Applied QLoRA** parameter-efficient fine-tuning to LLaMA 3.2 and Qwen 2.5 for domain-specific Python code generation; created prompt templates and evaluation harnesses measuring functional correctness of outputs; built structured data workflows for training and validation pipelines.
- Taught weekly lab sessions for **60+ graduate students** across Data Analytics, Python, SQL, and ML; produced course materials, structured learning modules, and assessment rubrics.

NISUM TECHNOLOGIES, INC.

Hyderabad, India

Senior Data Engineer

Nov 2018 – Dec 2023

Client: Williams-Sonoma, Inc. (625+ stores, 60+ countries)

Owined data engineering across 5 concurrent enterprise projects. Primary liaison between WSI business stakeholders and engineering on architecture, requirements, and delivery timelines.

- **Built 100+ ADF pipelines** and enhanced 60+ pipelines across Bronze-Silver-Gold Delta Lake layers in Azure Databricks; tuned PySpark and Scala jobs with partition pruning, Z-Ordering, and broadcast joins. **Reduced pipeline execution time by 40%.**
- **Co-architected** Snowflake Data Lakehouse consolidating **12 TB+** from Oracle, Teradata, Kafka, NetSuite ERP, and SFTP into ADLS Gen2 with dimensional and star schema modeling, SCD Type 2, clustering keys, virtual warehouse sizing, and Zero-Copy Cloning. **Cut infrastructure costs by 25%.**
- **Delivered ECDD** (customer-facing delivery date model) pipelines: implemented SCD Type 2 for carrier and routing dimensions, SQL date calculation and exception-handling logic across Global Reporting and B2B domains. **Reduced delivery inaccuracies by 30%.**
- **Led Oracle-to-Snowflake migration** (WMOS): designed ADF orchestration, built Databricks staging pipelines to Azure Blob, authored Scala Snowflake load scripts, and wrote SQL window functions for metric computation; profiled 20+ source tables and produced source-to-target mapping documents.
- **Architected centralized PIM system** as single source of truth across eCommerce and business systems; built ADF pipelines from Sterling OracleDB, SAP, and SFTP into ADLS/Blob staging across Orders, Products, Stores, Vendors, and Transportation domains.
- **Built and optimized Mi9 Retail Analytics pipelines:** ingested structured and semi-structured data (JSON, ORC, Parquet) from on-premises and cloud sources into Databricks; enabled Snowflake reporting cubes for sales analytics with full and incremental load patterns.
- **Drove Tableau-to-Power BI migration:** gathered ETL requirements from business stakeholders, authored data flow mapping documents, implemented business logic in ADF, and monitored daily and weekly batch runs for SLA compliance and data accuracy.
- **Established data quality frameworks:** schema validation, null-propagation detection, SCD deduplication logic, and source-to-target reconciliation across enterprise reporting domains. Managed CI/CD via Azure DevOps; integrated Azure Key Vault for secrets management; enforced least-privilege access.
- **Curated feature-ready datasets** for the ML team and coordinated data supply into Azure ML workflows; mentored **7 junior engineers** on pipeline standards and debugging. Maintained **100% sprint delivery** across quarterly Agile releases.

Client: Macy's, Inc. — Early-career data engineer supporting enterprise retail data warehouse.

- **Monitored 25+ ADF pipelines** across daily batch windows; ran source-to-target reconciliation to detect data loss and duplication; isolated root causes and coordinated fixes with senior architects.
- **Profiled 20+ operational tables** in Oracle and SQL Server (record counts, null distributions, schema alignment) to certify data readiness; improved complex SQL queries and resolved production warehouse issues including historical reloads.

PERSONAL PROJECTS

JOB AI MATCHER — AI-POWERED SEMANTIC JOB INTELLIGENCE PLATFORM

Python, FastAPI, PostgreSQL, PGvector, OpenAI Embeddings, GPT-4o-mini, Next.js, TypeScript, Supabase, Vercel, Render, JWT Auth

- **Developed and shipped** production AI job search platform: semantic resume matching via OpenAI embeddings and PGvector similarity search, real-time job ingestion from multiple APIs, GPT-4o-mini auto-apply with tailored cover letter generation, AI interview prep, application tracking dashboard, and full JWT + Argon2 auth. FastAPI backend on Render, Next.js on Vercel, PostgreSQL on Supabase.

EDUCATION

INDIANA UNIVERSITY

Indianapolis, Indiana

M.S. in Applied Data Science | GPA: 3.97 / 4.0

Aug 2023 – Dec 2025

Relevant Coursework: Statistical Learning, Database Design, Cloud Computing, Natural Language Processing, Computer Vision, Data Visualization

SKILLS

- **DATA ENGINEERING** — ETL/ELT Orchestration, Lakehouse Architecture, Dimensional & Star Schema Modeling, SCD Type 1/2, Data Quality & Governance, Data Lineage, Source-to-Target Reconciliation, CI/CD, DataOps
- **CLOUD & BIG DATA** — Azure (ADF, Databricks, ADLS Gen2, Blob Storage, Azure ML, Key Vault, Azure DevOps), Snowflake (Zero-Copy Cloning, Time Travel), PySpark, Spark SQL, Delta Lake, Kafka, Airflow, Oracle, SQL Server, Teradata
- **AI & MACHINE LEARNING** — LLM Fine-tuning (QLoRA), pgvector, OpenAI Embeddings API, NLP, Time Series Forecasting (ARIMA, Holt-Winters), Feature Engineering, scikit-learn
- **PROGRAMMING & TOOLS** — Python, SQL, Scala, PL/SQL, T-SQL, Penandas, NumPy, Power BI, FastAPI, SQLAlchemy, Next.js, Git, Jira, Confluence
- **CERTIFICATIONS** — Snowflake SnowPro Core (In Progress), Azure Data Engineer Associate DP-203 (In Progress), GenAI 101 (Indiana University)